Arithmetic & Number Sense

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Powers and roots

Evaluate $2^3 \times 2^2$ using the laws of exponents.

2 [Foundational] • Target: Working with fractions

Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?

3 [Foundational] • Target: Place value and order of magnitude

Write $4{,}500{,}000$ in the form $a \times 10^n$ with $1 \le a < 10$.

4 [Foundational] • Target: Place value and order of magnitude

Which is greater, 6×10^4 or $58{,}000$?

5 [Foundational] • Target: Place value and order of magnitude

Order from least to greatest: 0.6, $\frac{2}{3}$, 0.65.

6 [Foundational] • Target: Mixed numbers and improper fractions

Convert $4\frac{2}{5}$ to an improper fraction.

7 [Foundational] • Target: Mixed numbers and improper fractions

Convert $\frac{47}{6}$ to a mixed number.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Factors, multiples, GCD and LCM

Find the greatest common divisor (GCD) of 48 and 60.

9 [Intermediate] • Target: Factors, multiples, GCD and LCM

Find the least common multiple (LCM) of 8 and 12.

10 [Intermediate] • Target: Factors, multiples, GCD and LCM

List all the factors of 36, and state how many there are.

11 [Intermediate] • Target: Factors, multiples, GCD and LCM

Is 91 a prime number? Justify your answer.

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12 [Intermediate] • Target: Factors, multiples, GCD and LCM

Find the greatest common factor of 84 and 126.

13 [Intermediate] • Target: Divisibility rules

Without dividing, determine whether $4{,}578$ is divisible by 3.

14 [Intermediate] • Target: Divisibility rules

Is $2{,}340$ divisible by 9?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Estimation and rounding • Cross-Lesson: 3.01 Algebra Essentials

Estimate 198×51 by rounding each number to the nearest ten, then find the exact product.

16 [Challenging] • Target: Estimation and rounding • Cross-Lesson: 4.01 Geometry & Measurement

Round 2.4863 to two decimal places.

17 [Challenging] • Target: Estimation and rounding • Cross-Lesson: 5.01 Quantitative Comparisons

Estimate $\sqrt{50}$ to the nearest whole number.

18 [Challenging] • Target: Estimation and rounding • Cross-Lesson: 6.01 Sequences & Patterns

A shipment weighs approximately $2{,}850$ kg. Round this to the nearest hundred kilograms.

19 [Challenging] • Target: Number properties • Cross-Lesson: 7.01 Data Analysis & Probability

If n is an even integer, is $n^2 + n$ always even, always odd, or can it be either?

20 [Challenging] • Target: Number properties • Cross-Lesson: 8.01 Word Problems & Reasoning

Find the remainder when 47 is divided by 6.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Exponents and roots

Use the laws of exponents to evaluate $3^2 \times 3^4$.

a. 243

b. 81

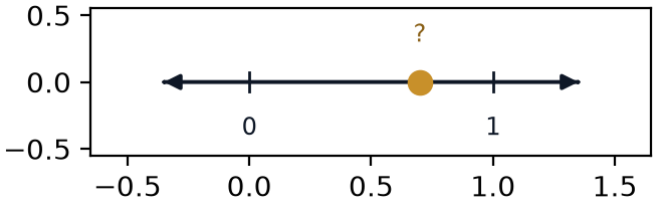
c. 729

d. 27

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22 [Foundational] • Target: MCQ — Comparing fractions and decimals

Which value is largest: the point shown on the number line, $\frac{3}{5}$, or 0.55?



a. The marked point (0.7)

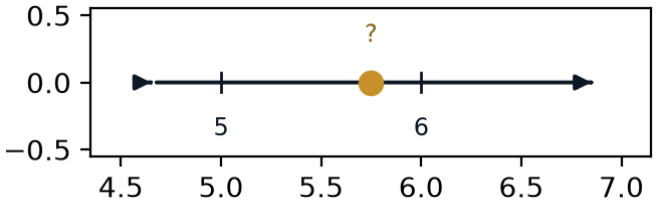
b. $\frac{3}{5}$

c. They are all equal

d. 0.55

23 [Foundational] • Target: MCQ — Decimal operations

The number line marks the value of $9.2 - 3.45$. Where does it land?



a. 5.75

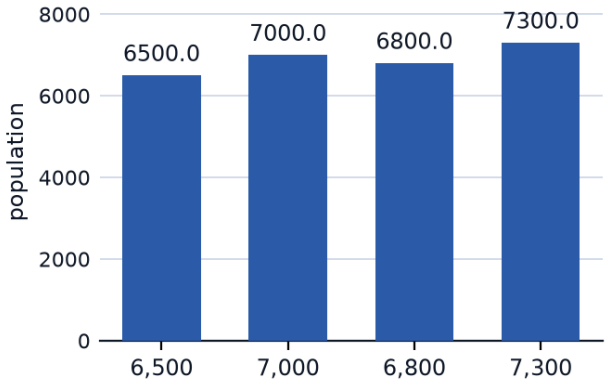
b. 6.25

c. 5.85

d. 5.65

24 [Foundational] • Target: MCQ — Scientific notation & place value

The bar chart shows four city populations. Which one equals 7×10^3?



a. 7,000

b. 6,800

c. 6,500

d. 7,300

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25 [Foundational] • Target: MCQ — Scientific notation & place value

Evaluate 3.4×10^4 as an ordinary number.

a. 3,400,000

b. 3,400

c. 340,000

d. 34,000

26 [Foundational] • Target: MCQ — Scientific notation & place value

In the number $528{,}437$, what is the place value of the digit 8?

a. Hundred thousands

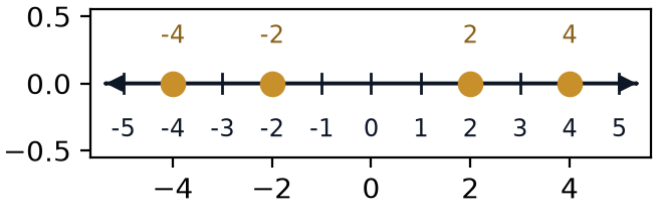
b. Thousands

c. Ten thousands

d. Hundreds

27 [Foundational] • Target: MCQ — Absolute value & signed numbers

Two of the four points on the number line represent a pair of numbers that are equal in absolute value but opposite in sign. Which pair?



a. -2 and 2

b. -4 and 4

c. -2 and 4

d. -4 and 2

28 [Foundational] • Target: MCQ — GCD, LCM & factors

How many factors does 60 have?

a. 6

b. 8

c. 12

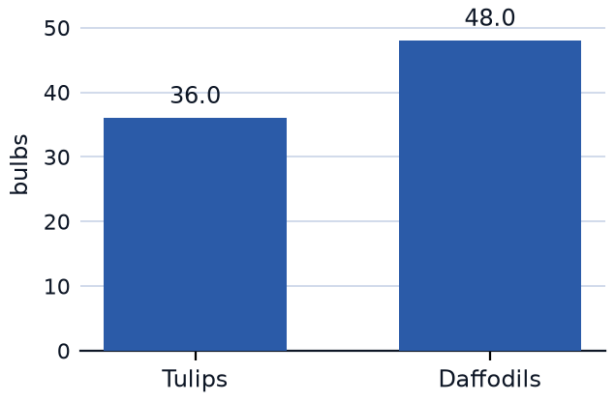
d. 10

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: MCQ — GCD, LCM & factors

The bar chart shows a gardener's bulb counts. They want to plant them in identical rows with the largest possible number of bulbs per row, each row containing only one type of flower. How many bulbs per row?



a. 4

b. 24

c. 6

d. 12

30 [Intermediate] • Target: MCQ — Divisibility rules

Without dividing, is $5{,}481$ divisible by 3?

a. Cannot tell without dividing

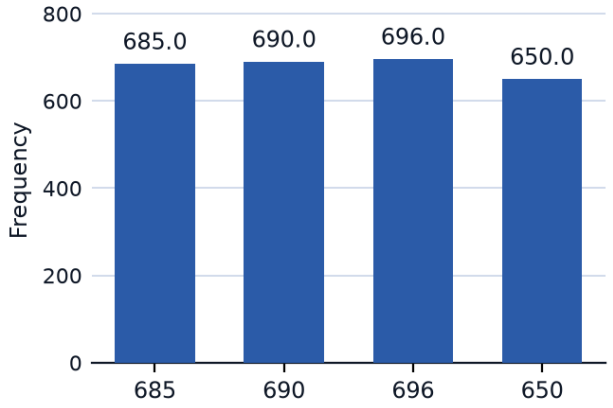
b. Yes

c. No

d. Only by 9, not 3

31 [Intermediate] • Target: MCQ — Divisibility rules

The bar chart shows four candidate numbers. Which one is divisible by both 5 and 6?



a. 690

b. 650

c. 685

d. 696

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32 [Intermediate] • Target: MCQ — Estimation & rounding

Round 147.362 to the nearest tenth.

a. 147.3

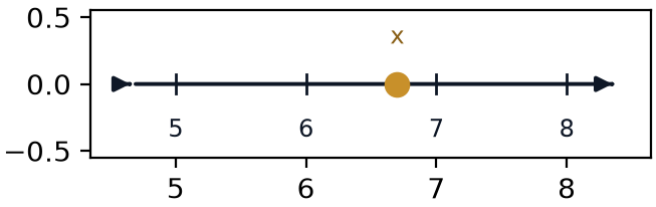
b. 150

c. 147.36

d. 147.4

33 [Intermediate] • Target: MCQ — Estimation & rounding

The number line shows the value x rounded to the nearest whole number. If $x = 6.7$, where does the rounded value land?



a. 6

b. 6.5

c. 8

d. 7

34 [Intermediate] • Target: MCQ — Estimation & rounding

Estimate 312×48 by rounding each factor to the nearest ten, then compare to the exact product $14{,}976$.

a. Estimate $310 \times 50 = 15{,}500$, far from the exact value

b. Estimate $300 \times 50 = 15{,}000$, close to the exact value

c. Estimate $320 \times 40 = 12{,}800$, close to the exact value

d. Estimate $310 \times 50 = 15{,}500$, close to the exact value

35 [Intermediate] • Target: MCQ — Number properties

If n is an odd integer, is $n^2 + n$ always even, always odd, or can it be either?

a. Can be either

b. Depends on the sign of n

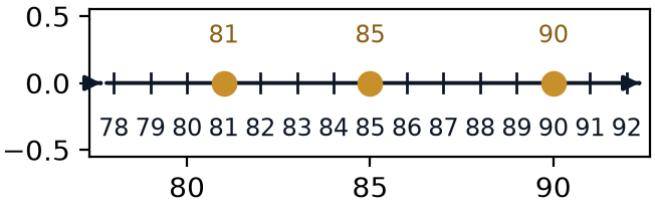
c. Always odd

d. Always even

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36 [Intermediate] • Target: MCQ — Number properties

The number line marks 85 between consecutive multiples of 9. Find the remainder when 85 is divided by 9.



a. 3

b. 4

c. 5

d. 76

37 [Intermediate] • Target: MCQ — Decimal operations

A shipment of 277 pencils is divided as equally as possible among 15 boxes. How many pencils does each box receive, and how many are left over?

a. $18, 7$

b. $18, 8$

c. $7, 18$

d. $19, 7$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Scientific notation & place value • Cross-Lesson: 6.01 Sequences & Patterns

A shipment of 310 books is divided as equally as possible among 15 shelves. How many books does each shelf receive, and how many are left over?

a. $20, 11$

b. $21, 10$

c. $10, 20$

d. $20, 10$

39 [Challenging] • Target: MCQ — Mixed numbers & improper fractions • Cross-Lesson: 7.01 Data Analysis & Probability

A shipment of 147 stickers is divided as equally as possible among 15 students. How many stickers does each student receive, and how many are left over?

a. $9, 12$

b. $9, 13$

c. $10, 12$

d. $12, 9$

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40 [Challenging] • Target: MCQ — Absolute value & signed numbers • Cross-Lesson: 8.01 Word Problems & Reasoning

A shipment of 336 chairs is divided as equally as possible among 23 rows. How many chairs does each row receive, and how many are left over?

a. 16, 13

b. 14, 14

c. 15, 14

d. 14, 15

41 [Challenging] • Target: MCQ — GCD, LCM & factors • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A shipment of 253 cups is divided as equally as possible among 14 bags. How many cups does each bag receive, and how many are left over?

a. 1, 18

b. 19, 1

c. 18, 2

d. 18, 1

42 [Challenging] • Target: MCQ — Divisibility rules • Cross-Lesson: 3.01 Algebra Essentials

[illegible] $x^2 - 6x + k = 0$[illegible] [illegible] k [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 6 [illegible] [illegible] [illegible] [illegible] [illegible] 5[illegible] [illegible] [illegible] [illegible] k [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] We have the equation $x^2 - 6x + k = 0$, where k is a constant. If the sum of the roots of this quadratic equation is 6 and the product of its roots is 5, what is the value of k, and what is the larger of the two roots?

a. $k = 5$[illegible] [illegible] [illegible] [illegible] 5

b. $k = 5$[illegible] [illegible] [illegible] [illegible] 1

c. $k = 6$[illegible] [illegible] [illegible] [illegible] 5

d. $k = 6$[illegible] [illegible] [illegible] [illegible] 1

43 [Challenging] • Target: MCQ — Estimation & rounding • Cross-Lesson: 4.01 Geometry & Measurement

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] How many 2-digit even numbers have distinct digits?

a. 40

b. 41

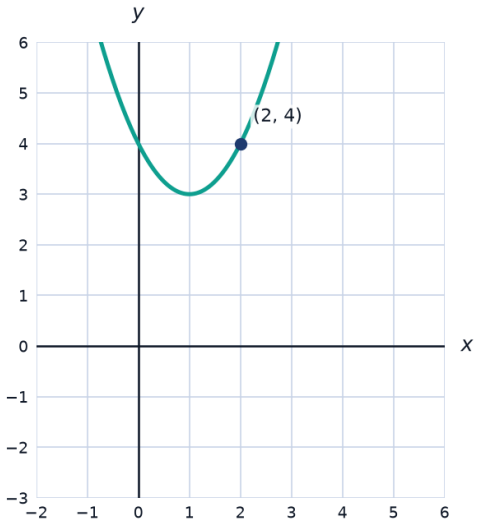
c. 45

d. 49

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44 [Challenging] • Target: MCQ — Number properties • Cross-Lesson: 5.01 Quantitative Comparisons

[redacted] $y = f(x)$. [redacted] $f(2)$[redacted] The graph above shows the function $y = f(x)$. What is the approximate value of $f(2)$?



a. 2

b. 3

c. 4

d. 5

45 [Challenging] • Target: Order of operations • Cross-Lesson: 6.01 Sequences & Patterns

[redacted] -2 [redacted] What is the number located at the mark -2 on the following number line?



a. -4

b. -1

c. -3

d. -2

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